THE TURNING WORLD

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Contents

1. Introduction 6
2. Notes 6
30th Oct, 2026: *Homeomorphism between the Cantor Set and* $\{0,1\}^{\mathbb{N}}$. 6
29th Oct, 2026: *Cantor Set.* 6
28th Oct, 2026: *Connected and Path Connected Components.* 6
27th Oct, 2026: *Connectedness in a Topological Space.* 7
26th Oct, 2026: *Clopen Subsets in a Topological Space.* 7
25th Oct, 2026: E_∞ *operad.* 7
24th Oct, 2026: *Commutative Topological Monoid.* 7
23rd Oct, 2026: *More on Infinite Symmetric Products.* 8
22nd Oct, 2026: *Extending Finite Symmetric Products.* 8
21st Oct, 2026: *Symmetric Products.* 8
20th Oct, 2026: *Operads.* 8
19th Oct, 2026: *Constant Tambara Functor.* 9
18th Oct, 2026: *Constant Mackey Functor.* 9
17th Oct, 2026: *Tambara Functor.* 9
16th Oct, 2026: *The double coset formula.* 10
15th Oct, 2026: *Conjugation defined as an automorphism.* 10
14th Oct, 2026: *Product of Projective Varieties.* 10
13th Oct, 2026: *Projective Space as a Completion of Affine Space.* 10
12th Oct, 2026: *Mackey Functor.* 11
11th Oct, 2026: *Limit points of a subset in a subspace.* 11
10th Oct, 2026: *Residue fields.* 12
09th Oct, 2026: *Localization.* 12
08th Oct, 2026: *Isomorphism of Affine Varieties.* 12
07th Oct, 2026: *Algebraic sets and radical ideals.* 13

06th Oct, 2026: *Viewing an Affine Variety through its ring of functions.* 13

05th Oct, 2026: *Zariski topology.* 14

04th Oct, 2026: *In continutation...* 14

03rd Oct, 2026: *The idea behind elementary Algebraic Geometry.* 14

02nd Oct, 2026: *Groups as Schemes.* 15

01st Oct, 2026: *Homology and Homotopy Groups.* 15

30th Sep, 2026: *Homotopy and Homology Groups.* 16

29th Sep, 2026: *Subsheaves and Subpresehaves.* 17

28th Sep, 2026: *Subobjects.* 17

27th Sep, 2026: *Quasi-projective varieties.* 17

26th Sep, 2026: *Projective Varieties.* 18

25th Sep, 2026: *Projective Space.* 18

24th Sep, 2026: *Homogeneous Ideal.* 18

23rd Sep, 2026: *Graded Ring.* 18

22nd Sep, 2026: *Quotient Space.* 18

21st Sep, 2026: *Action of a free group.* 19

20th Sep, 2026: *Action of $SL(2,\mathbb{Z})$.* 19

18th Sep, 2026: *The Category of Sets.* 20

17th Sep, 2026: *Isometric Group.* 21

16th Sep, 2026: *Tiling, mathematically.* 21

15th Sep, 2026: *Tiling of a Graph.* 21

14th Sep, 2026: *Farey Tree.* 21

13th Sep, 2026: *Drinfeld Center.* 22

11th Sep, 2026: *Affine Scheme and Scheme.* 22

10th Sep, 2026: *Symmetric Monoidal Categories.* 23

9th Sep, 2026: *Monoidal Categories.* 24

8th Sep, 2026: *Hilbert Nullstellensatz(s).* 24

7th Sep, 2026: *Understanding Tannaka.* 25

6th Sep, 2026: *The Farey Graph.* 25
5th Sep, 2026: *Center of a category.* 25
4th Sep, 2026: *More on germs and rational maps.* 26
3rd Sep, 2026: *On rational maps* 26
2nd Sep, 2026: *On regular morphisms* 27
1st Sep, 2026: *Subsheaves and Subpresehaves.* 28
31st Aug, 2026: *The action of Free groups.* 28
30th Aug, 2026 28
29th Aug, 2026 28
28th Aug, 2026 29
27th Aug, 2026 30
26th Aug, 2026 30
25th Aug, 2026 31
24th Aug, 2026 31
23rd Aug, 2026 33
22nd Aug, 2026 33
21st Aug, 2026 34
20th Aug, 2026 34
19th Aug, 2026 35
18th Aug, 2026 36
17th Aug, 2026 36
16th Aug, 2026 37
15th Aug, 2026 38
14th Aug, 2026 39
13th Aug, 2026 39
12th Aug, 2026 40
10th Aug, 2026 41
09th Aug, 2026 41

08th Aug, 2026 42
07th Aug, 2026 42
05th Aug, 2026 43
04th Aug, 2026 43
03rd Aug, 2026 44

1. Introduction

The motivation behind this running record of reading notes on mathematics is to gather the ongoing observations, ideas and questions which arise during my study. The entries are primarily for my own understanding, but I hope they may also be of interest to others who enjoy exploring mathematics. As my understanding evolves, some notes may be revised in the future.

2. Notes

30th Oct, 2026: ***Homeomorphism between the Cantor Set and*** $\{0,1\}^{\mathbb{N}}$***.*** The Cantor set C is homeomorphic to the product space $\{0,1\}^{\mathbb{N}}$, equipped with the product topology. The homeomorphism $\phi : \{0,1\}^{\mathbb{N}} \to C$ is defined by

$$\phi((a_n)_{n\geq 1}) = \sum_{n=1}^{\infty} \frac{2a_n}{3^n}, \tag{1}$$

where $a_n \in \{0,1\}$ for every $n \in \mathbb{N}$. This map replaces each binary digit a_n with the ternary digit $2a_n$, thereby representing every point of the Cantor set using a ternary expansion containing only 0 and 2. The map is bijective and continuous, and its inverse is continuous.

29th Oct, 2026: ***Cantor Set.*** The Cantor set is a fundamental example in topology because it exhibits several seemingly contrasting properties simultaneously. It is compact, perfect, uncountable, totally disconnected, and nowhere dense in $\mathbb{R}$. In particular, its connected components are singletons, despite the set being uncountable. Moreover, it is homeomorphic to the countable product $\{0,1\}^{\mathbb{N}}$, providing a connection between topology and infinite binary sequences. It also admits an abelian group structure under coordinate-wise addition modulo 2. These properties make the Cantor set a useful model for studying disconnected compact spaces and the interplay between topology and algebra.

28th Oct, 2026: ***Connected and Path Connected Components.*** Every path-connected component of a topological space is contained in a connected component, since path-connectedness implies connectedness. In general, a connected component may contain one or more path-connected components. If every connected component is path-connected, then the connected components and path-connected components coincide. This holds, in particular, for locally path-connected spaces. If a connected component is not path-connected, it contains multiple path-connected components, so the two types of components do not coincide entirely. Thus, the components always coincide when the space is locally path-connected, but in arbitrary spaces, they may coincide partially or not at all.

27th Oct, 2026: ***Connectedness in a Topological Space.*** Let A be a connected subspace of a topological space X. Then its closure $\overline{A}$ is also connected.

Suppose that $\overline{A}$ admits a separation into two nonempty, disjoint relatively open subsets. Since A is dense in its closure, both pieces cannot be separated from A without contradicting the density of A. Consequently, such a separation would induce a separation of A, contradicting its connectedness.

26th Oct, 2026: ***Clopen Subsets in a Topological Space.*** A topological space is disconnected if it can be expressed as the union of two disjoint, nonempty open subsets. Since these subsets are complements of one another, they are also closed. Thus, a space is connected when its only clopen subsets are the empty set and the entire space.

This gives us a useful way to think about connectedness, i.e., a connected space cannot be separated into two nonempty pieces that are simultaneously open and closed.

25th Oct, 2026: E_∞ ***operad.*** We have seen that the infinite symmetric product $\mathrm{SP}^\infty(X)$ acquires the structure of a commutative topological monoid. However, in homotopy theory, we often want to study structures that are commutative and associative not necessarily strictly, but up to coherent homotopies. This motivates the notion of an E_∞ space.

An E_∞ operad $\mathcal{E}$ is an operad such that each space $\mathcal{E}(n)$ is contractible and carries a free action of the symmetric group Σ_n. An E_∞ space is a topological space X equipped with an action of such an operad, given by map

$$\mathcal{E}(n) \times X^n \longrightarrow X \tag{2}$$

compatible with operadic composition and symmetric group actions.

The contractibility of $\mathcal{E}(n)$ ensures that the n-ary operations are unique up to homotopy, while the operadic structure organises these homotopies coherently. Thus, an E_∞ space captures commutative algebraic structure together with the higher homotopical information that strict commutative monoids may fail to retain.

24th Oct, 2026: ***Commutative Topological Monoid.*** Let X be a based space with basepoint x_0, and let $\mathrm{SP}^\infty(X)$ be its infinite symmetric product.

For any commutative topological monoid M and any continuous based map $f : X \to M$, there exists a unique continuous monoid homomorphism

$$\bar{f} : \mathrm{SP}^\infty(X) \to M \tag{3}$$

such that

$$\bar{f}([x_1, \ldots, x_n]) = f(x_1) \cdots f(x_n). \tag{4}$$

The basepoint must map to the monoid identity, so inserting or removing x_0 does not change the product.

23rd Oct, 2026: ***More on Infinite Symmetric Products.*** We can combine two collections of points in $\mathrm{SP}^\infty(X)$ by taking their union, making $\mathrm{SP}^\infty(X)$ a commutative topological monoid.

Categorically, this construction has a universal property: it is the free commutative topological monoid generated by X (under suitable assumptions on the space). Thus, maps from X into a commutative topological monoid extend uniquely to monoid maps from $\mathrm{SP}^\infty(X)$.

22nd Oct, 2026: ***Extending Finite Symmetric Products.*** Having defined the finite symmetric product $\mathrm{SP}^n(X)$, we can ask whether there is a natural way to let the number of points vary without bound. To do this, we first choose a basepoint $x_0 \in X$. We then define maps

$$\mathrm{SP}^n(X) \longrightarrow \mathrm{SP}^{n+1}(X) \tag{5}$$

by sending an unordered collection $[x_1, \ldots, x_n]$ to $[x_1, \ldots, x_n, x_0]$.

We essentially enlarge the collection by adding the basepoint. The infinite symmetric product is then the colimit

$$\mathrm{SP}^\infty(X) = \operatorname*{colim}_n \mathrm{SP}^n(X). \tag{6}$$

The topology is inherited through this colimit construction. Thus, we pass from studying unordered collections of a fixed size to collections of arbitrary finite size, with the basepoint providing the natural inclusion between successive stages.

21st Oct, 2026: ***Symmetric Products.*** Suppose we have a space X and want to study collections of n points in X without distinguishing between their order. The natural construction is to take the Cartesian product X^n and quotient by the action of the symmetric group (S_n, however here we call it) Σ_n, which permutes the coordinates. We obtain the nth symmetric product as

$$\mathrm{SP}^n(X) = X^n/\Sigma_n. \tag{7}$$

Thus, a point in $\mathrm{SP}^n(X)$ is an unordered collection of n points, with repetitions allowed. The philosophy is to retain the information about which points occur while forgetting the order in which they are listed.

20th Oct, 2026: ***Operads.*** An associative algebra has a binary multiplication which can be composed with itself to produce operations of higher arity. An operad provides a framework for describing such compositions abstractly.

A (symmetric) operad $\mathcal{O}$ consists of a collection of objects $\mathcal{O}(n)$, one for each $n \geq 0$, together with composition maps

$$\mathcal{O}(k) \otimes \mathcal{O}(n_1) \otimes \cdots \otimes \mathcal{O}(n_k) \longrightarrow \mathcal{O}(n_1 + \cdots + n_k).$$

Intuitively, an element of $\mathcal{O}(k)$ is a k-input operation, and composition amounts to inserting other operations into its inputs. These compositions must satisfy associativity and

unit axioms, along with compatibility with the symmetric group actions that permute the inputs.

19th Oct, 2026: ***Constant Tambara Functor.*** A simple example to consider is the constant Tambara functor associated with the group $C_2 = \{e, g\}$. Let us take $\mathbb{Z}$ as the value at both subgroups, e and C_2. The restriction map is the identity, while transfer and norm are given by

$$\mathrm{tr}_e^{C_2}(n) = 2n, \qquad N_e^{C_2}(n) = n^2. \tag{8}$$

The norm is multiplicative but not additive. And,

$$N(a+b) = a^2 + 2ab + b^2 = N(a) + \mathrm{tr}(ab) + N(b).$$

which illustrates the distributivity relation between norm and transfer, and this in turn distinguishes Tambara functors from Mackey functors.

18th Oct, 2026: ***Constant Mackey Functor.*** A simpler example to begin with is the constant Mackey functor. Let G be a finite group and let A be an abelian group. We associate the same abelian group A to every subgroup $H \leq G$. For an inclusion $K \leq H$, we define the restriction map to be the identity, while the transfer map is defined to be the multiplication by the index $[H]$:

$$\mathrm{res}_K^H = \mathrm{id}_A, \qquad \mathrm{tr}_K^H(x) = [H]x. \tag{9}$$

All conjugation maps are also taken to be the identity on A. These maps satisfy the Mackey functor axioms. In particular, the double coset formula reduces to a counting identity for the orbits of H acting on G/K.

17th Oct, 2026: ***Tambara Functor.*** Having looked at Mackey functors, a natural question is whether the structure of a Mackey functor captures all the operations one might want to study in equivariant algebra. A Mackey functor provides restriction and additive transfer maps, together with the compatibility conditions relating them. But we also wish to know what happens when we also want to encode multiplicative norms?

A *Tambara* functor extends the structure of a Mackey functor by incorporating norm maps alongside restriction and transfer.

For a map of finite G-sets $f : X \to Y$, these operations are conventionally written as

$$f^* : T(Y) \to T(X), \qquad f_+ : T(X) \to T(Y), \qquad f_\bullet : T(X) \to T(Y). \tag{10}$$

Here, f^* is restriction, f_+ is additive transfer, and $f_\bullet$ is the multiplicative norm. Unlike transfer, a norm is generally not additive. Consequently, one cannot simply treat it as another homomorphism of abelian groups.

This raises the central question: how should norms interact with addition and transfer? And the answer is given by the distributivity axioms, expressed through exponential diagrams. These axioms ensure that the three operations fit together coherently.

16th Oct, 2026: ***The double coset formula.*** In the definition of a Mackey functor, we have restriction and transfer maps between groups associated to different subgroups. A natural question to ask is: what happens when we first transfer an element from a subgroup $K \leq G$ to G and then restrict it to another subgroup $H \leq G$?

These operations are defined separately, but their composite is governed by the double coset formula $\sum_{[g]\in H\backslash G/K} \operatorname{tr} {}^{H}_{H\cap gKg^{-1}} \circ c_g \circ \operatorname{res}^{K}_{g^{-1}Hg\cap K}$.

When we restrict the action of G on G/K to H, the resulting H-orbits are indexed by $H\backslash(G/K)$. The stabiliser of the orbit represented by gK is $H \cap gKg^{-1}$. Thus, each term in the formula corresponds to one orbit, with restriction, conjugation and transfer accounting for its contribution.

For $G = C_p$ and $H = K = e$, the formula reduces to

$$\operatorname{res}^{C_p}_{e} \circ \operatorname{tr} e^{C_p}(x) = \sum i = e^{p-1}c^i(x). \tag{11}$$

15th Oct, 2026: ***Conjugation defined as an automorphism.*** When we look at the definition of a C_p-Mackey functor, a natural question to ask is why the conjugation map is required to be an automorphism of order dividing p.

Recall that $C_p = \langle g \rangle$ acts on the underlying abelian group $M(C_p/e)$ through the conjugation map c. Since $g^p = e$, the action must satisfy $c^p = \mathrm{id}$. Hence the conjugation map encodes an action of C_p on the underlying group.

The other axioms require $c \circ \mathrm{res} = \mathrm{res}$ and $\mathrm{tr} \circ c = \mathrm{tr}$. Moreover, the identity connects conjugation with restriction and transfer: $\mathrm{res}(\mathrm{tr}(x)) = \sum_{i=0}^{p-1} c^i(x)$.

14th Oct, 2026: ***Product of Projective Varieties.*** The product $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$ are both two-dimensional projective varieties, but they are not isomorphic. A point of $\mathbb{P}^2$ is given by a single triple of homogeneous coordinates, whereas a point of $\mathbb{P}^1 \times \mathbb{P}^1$ is a pair of points, each belonging to a projective line.

The product can also be embedded into $\mathbb{P}^3$ as the quadric surface $XW - YZ = 0$, through the Segre embedding. This gives a geometric way to see the product inside a higher-dimensional projective space and illustrates that varieties of the same dimension can have different geometric structures.

13th Oct, 2026: ***Projective Space as a Completion of Affine Space.*** Projective space extends affine space by encoding directions as points. In $\mathbb{P}^2(k)$, the points with

nonzero last coordinate can be written as $[x : y : 1]$, giving an affine plane $\mathbb{A}^2(k)$. The remaining points, of the form $[x : y : 0]$, constitute the line at infinity.

Each point on this line represents a direction in the affine plane. Consequently, parallel affine lines intersect at the point corresponding to their common direction when extended projectively. Homogeneous polynomials define equations in the projective space that remain unchanged under rescaling of homogeneous coordinates.

12th Oct, 2026: ***Mackey Functor.*** Let C_p be a cyclic group of prime order. Then a C_p-Mackey functor is the data of 2 abelian groups $M(C_p/e)$ (the underlying) and $M(C_p/C_p)$ (the fixed), with three abelian group homomorphisms: namely, restriction, trasnfer, and conjugation:

$$\text{res} : M(C_p/C_p) \to M(C_p/e), \text{tr} : M(C_p/e) \to M(C_p/C_p), \text{conj} : M(C_p/e) \to M(C_p/e) \tag{12}$$

which satisfy the following axioms:

(1) conj is an automorphism of order dividing p,
(2) $\text{conj} \circ \text{res} = \text{res}$ and $\text{tr} \circ \text{conj} = \text{tr}$,
(3) $\text{res}(\text{tr}(x)) = \sum_{i=1}^{p-1} \text{conj}^i(x)$, for every $x \in M(C_p/e)$.

If we consider both the abelian groups to be the trivial group, then all the maps are trivial and we call this the *zero* Mackey functor.

We can also define a Mackey functor homomorphims, given two Mackey functors. We could perhaps discuss that in the next note.

11th Oct, 2026: ***Limit points of a subset in a subspace.*** Given a space X , a subspace Y , and a subset $A \subset Y$, the closure of A in the space Y is the intersection of the closure of A in X with Y.

($\Longrightarrow$) Let $x \in \text{Cl}_Y(A)$. By definition, $\text{Cl}_Y(A)$ is the intersection of all closed sets in Y that contain A.

Now, $\text{Cl}_X(A)$ is a closed subset of X containing A. Then, in the subspace topology of Y, $\text{Cl}_X(A) \cap Y$ is closed in Y and contains A.

Since $x \in \text{Cl}_Y(A)$, hence $x \in \text{Cl}_X(A) \cup Y$.

Thus, $\text{Cl}_Y(A) \subseteq \text{Cl}_X(A)$.

($\Longleftarrow$) Let $x \in \text{Cl}_X(A) \cap Y$. Hence, $x \in \text{Cl}_X(A)$ and $x \in Y$.

Let us assume that $x \notin \text{Cl}_Y(A)$. Then $x \in Y \setminus \text{Cl}_Y(A)$, which is an open set.

By the definition of the subspace topology, any open set in Y can be written as $U \cap Y$ for some open set U in X. Thus, there is an open set U in X such that $x \in U \cap Y \subseteq Y \setminus \text{Cl}_Y(A)$.

In particular, since $A \subseteq \mathrm{Cl}_Y(A)$, the set U is disjoint from A.

Consequently, $X \setminus U$ is a closed subset of X that contains A. Since $x \in \mathrm{Cl}_X(A)$ and $\mathrm{Cl}_X(A)$ is the intersection of all closed sets in X containing A, x must belong to $X \setminus U$, meaning $x \notin U$.

This contradicts our earlier deduction that $x \in U \cap Y$. Therefore, our assumption that $x \notin \mathrm{Cl}_Y(A)$ must be false.

Thus, $\mathrm{Cl}_X(A) \cap Y \subseteq \mathrm{Cl}_Y(A)$.

10th Oct, 2026: ***Residue fields.*** Let A be a commutative ring and $\mathfrak{p} \in \mathrm{Spec}(A)$. The residue field of $\mathfrak{p}$ is defined as

$$\kappa(\mathfrak{p}) := A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}. \tag{13}$$

Since $A_{\mathfrak{p}}$ is a local ring with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$, this quotient is a field.

Moreover, $\kappa(\mathfrak{p}) \cong \mathrm{Frac}(A/\mathfrak{p})$.

Geometrically, the residue field captures the field of values associated with a point. For example, if $A = k[x]$ and $\mathfrak{p} = (x - a)$, where $a \in k$, then $\kappa(\mathfrak{p}) \cong k$. However, for $A = \mathbb{R}[x]$ and $\mathfrak{p} = (x^2 + 1)$, we obtain $\kappa(\mathfrak{p}) \cong \mathbb{C}$.

More generally, for a scheme X and $x \in X$, we define $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$.

Residue fields connect the local algebra of a scheme with the geometry of its points.

09th Oct, 2026: ***Localization.*** Suppose X is an affine variety and $f \in k[X]$. We can consider the subset $D(f) = \{p \in X : f(p) \neq 0\}$.

These are the basic open sets in the Zariski topology. Here, instead of imposing an equation, we are restricting to the points where a particular polynomial does not vanish.

We want to know what happens algebraically when we restrict to $D(f)$? We want to allow f to behave like an invertible element, since it never vanishes on this set. This then motivates the localization $k[X]_f = k[X]\left[\frac{1}{f}\right]$.

For example, on the affine line, $D(x)$ is the line with the origin removed. Its coordinate functions can involve $1/x$, so the corresponding ring is $k[x, x^{-1}]$.

We essentially observe that removing the zero set of a polynomial geometrically corresponds to making that polynomial invertible algebraically. Localization lets us study a variety on smaller open pieces, which becomes important when we move from affine varieties to schemes.

08th Oct, 2026: ***Isomorphism of Affine Varieties.*** Two affine varieties X and Y are isomorphic if there are regular maps $F : X \to Y$ and $G : Y \to X$ such that $G \circ F = \mathrm{id}_X$

and $F \circ G = \mathrm{id}_Y$. That is, they are the same as varieties, even if they sit inside different affine spaces.

For example, consider the parabola $X = V(y - x^2) \subseteq \mathbb{A}^2$. The map $F : \mathbb{A}^1 \to X, \qquad t \mapsto (t, t^2)$ has inverse given by projection onto the first coordinate, $(x, y) \mapsto x$. Hence $X \cong \mathbb{A}^1$.

This is also reflected in their coordinate rings $k[X] \cong k[x]$.

So the parabola is embedded in the plane, but as an affine variety it is isomorphic to the affine line.

07th Oct, 2026: ***Algebraic sets and radical ideals.*** We have seen that an ideal I gives an algebraic set $V(I)$, but that different ideals can have the same zero set. The simplest example is $V(x) = V(x^2)$ over an algebraically closed field.

The natural question is therefore: given $V(I)$, what is the largest ideal that defines exactly the same set?

We already have a candidate $I(V(I)) = \{f : f(a) = 0 \text{ for every } a \in V(I)\}$.

Hilbert's Nullstellensatz tells us that $I(V(I)) = \sqrt{I}$. So passing from I to $V(I)$ loses the information contained in the non-radical part of I.

For example, $\sqrt{(x^2)} = (x)$, which explains why $x = 0$ and $x^2 = 0$ define the same algebraic set.

This makes the earlier correspondence between algebraic sets and radial ideals quite precise. Also, when the algebraic set is irreducible, the corresponding radical ideal is prime.

06th Oct, 2026: ***Viewing an Affine Variety through its ring of functions.*** For an affine variety $X \subseteq k^n$, we have already seen that the ideal $I(X) = \{f \in k[x_1, \ldots, x_n] : f(a) = 0 \text{ for all } a \in X\}$ captures the polynomial equations satisfied by all points of X.

We can go one step further and consider $k[X] = k[x_1, \ldots, x_n]/I(X)$.

This is the coordinate ring of X. We take the quotient, following the intuition that two polynomials which agree on X should be regarded as the same function on X. Their difference then lies in $I(X)$.

For example, if $X = V(y - x^2)$, then $y - x^2 = 0$ on X, so in the coordinate ring we simply have $y = x^2$.

Hence $k[X] = k[x, y]/(y - x^2) \cong k[x]$.

So even though X sits inside $\mathbb{A}^2$, its coordinate ring tells us that its polynomial functions behave like polynomials in one variable. This is the beginning of viewing an affine variety through its ring of functions rather than only through its points.

05th Oct, 2026: ***Zariski topology.*** We have seen that polynomial equations naturally give subsets of k^n. For an ideal $I \subseteq k[x_1, \ldots, x_n]$, $V(I) = \{a \in k^n : f(a) = 0 \text{ for every } f \in I\}$.

We can use these algebraic sets as the closed sets of a topology on k^n, called the *Zariski topology*.

The topology axioms follow from the algebra of ideals:

(1) $\bigcap_\alpha V(I_\alpha) = V\left(\sum_\alpha I_\alpha\right)$, and
(2) $V(I) \cup V(J) = V(IJ)$.

If k is algebraically closed, the proper closed subsets of $\mathbb{A}^1_k$ are just finite sets, since a nonzero polynomial has only finitely many roots. Thus $\mathbb{A}^1_k \setminus \{a_1, \ldots, a_r\}$ is Zariski-open.

There is also a direct connection with the earlier discussion of ideals: $V(I) = V(\sqrt{I})$.

Thus the geometry here cannot really distinguish an ideal from its radical. Hence, one can begin to realise that algebraic geometry naturally involves a translation between algebraic properties of rings and geometric properties of (topological) spaces.

04th Oct, 2026: ***In continutation...*** Once we have the ideal generted by the polynomials in the space we wish to talk about, we then consider $V(I)$, which is simply the set of common zeros of everything in I.

Essentially, the geometry is attached to the algebraic structure generated by the polynomials.

However, different ideals can define the same algebraic set. In fact, if we start with an ideal I, take its zero set, and then ask for all polynomials vanishing on that zero set, we generally do not recover I, but its radical: $I(V(I)) = \sqrt{I}$ over an algebraically closed field. Here, the geometry does not preserve all the algebraic information.

The concept of irreducibility enters naturally. Consider $V(xy) = V(x) \cup V(y)$, so the equation $xy = 0$ gives two lines rather than one irreducible geometric piece. On the algebraic side, this corresponds to the ideal (xy) not being prime.

I think this is probably one of the things that makes algebraic geometry so appealing.

03rd Oct, 2026: ***The idea behind elementary Algebraic Geometry.*** I like how quickly algebraic geometry starts blurring the distinction between an equation and the geometric object it defines. We start with something as simple as $f(x_1, \ldots, x_n) = 0$, and suddenly we are thinking of its solutions as a geometric object.

For example, $y = x^2$ is just a polynomial equation in two variables, but its zero set is a parabola.

This becomes more interesting when there is more than one equation. Given polynomials $f_1, \ldots, f_r$, we look at $V(f_1, \ldots, f_r) = \{a \in k^n : f_1(a) = \cdots = f_r(a) = 0\}$.

Now the natural question becomes- what information about the equations matters for the geometry?

This is where ideals enter. If $f_1, \ldots, f_r$ all vanish at a point a, then every polynomial combination $g_1 f_1 + \cdots + g_r f_r$ also vanishes at a. So instead of keeping track of a particular list of equations, it makes more sense to consider the ideal $I = (f_1, \ldots, f_r)$.

02nd Oct, 2026: ***Groups as Schemes.*** Let k be a field. We know that a scheme is a locally ringed space $(X, \mathcal{O}_X)$ which is locally isomorphic to $\operatorname{Spec} A$ for some commutative ring A. In particular, an affine scheme is one of the form $X = \operatorname{Spec} A$, and the ring A is called the coordinate ring of X.

If we work with k-algebras, the category of affine k-schemes is equivalent to the opposite of the category of commutative k-algebras: $\mathbf{Aff}_k \simeq (\mathbf{CAlg}_k)^{\mathrm{op}}$.

The appearance of the opposite category comes from the fact that a ring homomorphism $A \longrightarrow B$ induces a morphism of affine schemes in the opposite direction, $\operatorname{Spec} B \longrightarrow \operatorname{Spec} A$.

Now suppose G is a group. We can regard G as a k-scheme by taking one copy of $\operatorname{Spec} k$ for every element of G: $\underline{G} = \coprod_{g \in G} \operatorname{Spec} k$. This is called the constant group scheme associated to G.

The group multiplication $G \times G \to G, (g, h) \longmapsto gh$ then gives a morphism of schemes

$$\underline{G} \times_k \underline{G} \longrightarrow \underline{G}.$$

Likewise, the identity element and the inverse map give morphisms

$$\operatorname{Spec} k \longrightarrow \underline{G}, \qquad \underline{G} \longrightarrow \underline{G}.$$

Hence the group axioms are realized as morphisms of schemes, so $\underline{G}$ is an affine group scheme.

Thus we obtain an embedding

$$\mathbf{Grp} \hookrightarrow \mathbf{AffGrp}_k, \qquad G \longmapsto \underline{G}.$$

01st Oct, 2026: ***Homology and Homotopy Groups.*** The fundamental group, i.e., the first homotopy group remembers how loops can be composed, and this composition need not be commutative. Homology, however, is built from an additive structure, so it necessarily loses some of this non-commutative information. Passing from $\pi_1(X)$ to $H_1(X)$ is the process of forcing the fundamental group to become abelian.

The difference becomes clearer when we look at higher dimensions. For a sphere S^n, homology is particularly simple:

$$H_k(S^n) \cong \begin{cases} \mathbb{Z}, & k = 0, \cdots, n, \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

One might therefore expect the homotopy groups to have a similarly simple description...

Let us consider $\pi_2(S^2) \cong \mathbb{Z}$, which agrees with $H_2(S^2) \cong \mathbb{Z}$. But then $\pi_3(S^2) \cong \mathbb{Z}$, while $H_3(S^2) = 0$. So there is non-trivial homotopy in dimension 3 even though there is no third homology group.

A map $S^3 \to S^2$ can represent a genuinely non-trivial homotopy class even though there is no 3-dimensional homology class in S^2 corresponding to it.

Nevertheless, the agreement between $\pi_2(S^2)$ and $H_2(S^2)$ is part of a much more general phenomenon. The Hurewicz theorem gives us a bridge between the two theories.

30th Sep, 2026: ***Homotopy and Homology Groups.*** Homotopy groups arise from thinking about maps from spheres into a space, while homology groups arise from chains, cycles, and boundaries. It is natural, then, to ask why these two constructions should be related at all.

For a pointed space X, the homotopy group $\pi_n(X)$ can be thought of, roughly, as the collection of maps

$$f : S^n \to X \tag{15}$$

where two maps are considered the same if one can be continuously deformed into the other.

Homology approaches the same space differently, where we construct groups of chains and use the boundary map

$$\partial_n : C_n(X) \to C_{n-1}(X). \tag{16}$$

An n-cycle is an element in $\ker \partial_n$, while an n-boundary is an element of $\operatorname{im} \partial_{n+1}$. The homology group is then

$$H_n(X) = \frac{\ker \partial_n}{\operatorname{im} \partial_{n+1}}. \tag{17}$$

We know that $\pi_1(S^1) \cong \mathbb{Z}$. The integer here records the winding number of a loop around the circle. Interestingly, the first homology group gives the same answer $H_1(S^1) \cong \mathbb{Z}$.

For a path-connected space X, there is a natural isomorphism

$$H_1(X; \mathbb{Z}) \cong \pi_1(X)^{\mathrm{ab}}, \tag{18}$$

where $\pi_1(X)^{\mathrm{ab}}$ is the abelianization of the fundamental group.

29th Sep, 2026: ***Subsheaves and Subpresehaves.*** Let F be a presheaf on a topological space X. A subpresheaf of F is a presheaf G together with a monomorphism

$$G \hookrightarrow F \tag{19}$$

in the category of presheaves on X. Thus for every open set $U \subseteq X$, we choose a subset $G(U) \subseteq F(U)$ such that the restriction maps of F restrict to maps between the chosen subsets. For every inclusion $V \subseteq U$, we require

$$\rho^F_{UV}\big(G(U)\big) \subseteq G(V). \tag{20}$$

where ρ denotes the restriction map.

Now, if F is a sheaf, a subsheaf of F is a sheaf G together with a monomorphism

$$G \hookrightarrow F \tag{21}$$

in the category $\mathrm{Sh}(X)$. Equivalently, we may regard $G(U)$ as a subset of $F(U)$ for every open set U, with the restriction maps of F restricting to those of G, and with G itself satisfying the sheaf axioms.

Thus, every subsheaf is in particular a subpresheaf, but a subpresheaf of a sheaf need not be a sheaf. The additional requirement is that the chosen sections satisfy the gluing and locality conditions of a sheaf.

28th Sep, 2026: ***Subobjects.*** Suppose X is an object in a category $\mathscr{C}$. A *subobject* of X is an equivalence class of monomorphisms

$$m : A \hookrightarrow X. \tag{22}$$

Two monomorphisms $m : A \hookrightarrow X$ and $n : B \hookrightarrow X$ are considered equivalent if there exists an isomorphism $f : A \to B$ such that

$$n \circ f = m. \tag{23}$$

In the category **Set**, monomorphisms are the injective maps, and hence subobjects of a set correspond to its subsets.

27th Sep, 2026: ***Quasi-projective varieties.*** Since we briefly touched upon quasi-projective varieties, let us define them. A quasi-projective variety is a locally closed subset of a projective variety.

In fact, an interesting thing to ponder on is that every affine variety is quasi-projective.

26th Sep, 2026: ***Projective Varieties.*** Now, a projective algebraic variety is an irreducible algebaric set in P_k^n with the induced topology.

The induced topology here is defined by the Zariski topology on P_k^n by taking open sets to be the complements of the (closed) algebaric sets.

Then, an open subset of a projective variety is a *quasi-projective variety.* And the dimension of a projective or a quasi-projective variety is its dimension as a topological space.

This direction of thinking and developign the theorey will eventually lead us to a variant of Hilbert's Nullstellansatz, which here (at least in Hartshorne) is referred ot as *homogeneous Nullstellansatz.*

25th Sep, 2026: ***Projective Space.*** Let k be an algebraically closed field. We define a projective n-space over k, denoted P_k^n to be the set of equivalence classes of $(n+1)$ tuples $(a_0, \cdots, a_n)$ of elements of k, not all zero, under the equivalence relation $(a_0, \cdots, a_n) \sim (b_0, \cdots, b_n)$ if there exists $\lambda \in k^*$ such that $a_i = \lambda b_i$ for all i.

We define a projective space because ultimately we are interested in projective varieties. A projective variety is a subset of $\mathbb{P}^n$ defined as the simultaneous vanishing set (or zero locus) of a collection of homogeneous polynomials. We will discuss more about them in later notes.

24th Sep, 2026: ***Homogeneous Ideal.*** Let $R = \bigoplus_{d \geq 0} R_d$ be a graded ring. We say that an ideal $I \subseteq R$ is *homogeneous* if it is compatible with the grading, in the sense that whenever $f = \sum_d f_d \in I$, with $f_d \in R_d$, then each homogeneous component f_d also belongs to I. Equivalently, I is generated by homogeneous elements.

23rd Sep, 2026: ***Graded Ring.*** A *grading* on a ring R is a decomposition of R as a direct sum of additive subgroups $R = \bigoplus_{d \geq 0} R_d$ such that $R_d R_e \subseteq R_{d+e}$ for all $d, e \geq 0$.

The elements of R_d are called *homogeneous elements of degree d.* Thus every element $f \in R$ can be written uniquely as a finite sum

$$f = f_0 + f_1 + \cdots + f_r, \qquad f_d \in R_d. \tag{24}$$

The condition $R_d R_e \subseteq R_{d+e}$ makes this decomposition compatible with multiplication.

22nd Sep, 2026: ***Quotient Space.*** Let X be a topological space and let $\sim$ be an equivalence relation on X. Then $Y = X/\sim$ is defined as the set of equivalence classes of elemenets in X, also known as the *quotient set.* This gives us a canonical surjection $q : X \to Y$ such that $x \mapsto [x]$.

The quotient space under $\sim$ is the Y equipped wiht the quotient topology, whose open sets are those subsets $U \subseteq Y$ whose pre-image $q^{-1}(U)$ is open.

21st Sep, 2026: ***Action of a free group.*** A beautiful theorem in GGT states that *If a group acts freely on a tree, then the group is free.* We will discuss the sketch of the proof of this theorem.

Suppose that G acts freely on a tree T. We want to show that G is free.

We choose a vertex $v \in T$. For every $g \in G$, consider the unique path from v to gv. Since T is a tree, this path is uniquely determined.

Now we choose a set of elements of G which move v to the vertices adjacent to v along these paths. We want to regard these elements as generators. The important point is to understand what happens when we multiply such elements. A word

$$g_1 g_2 \cdots g_n \tag{25}$$

determines a sequence of vertices

$$v, \quad g_1 v, \quad g_1 g_2 v, \quad \ldots, \quad g_1 g_2 \cdots g_n v. \tag{26}$$

Thus a word determines a path in the tree.

If the word is reduced, consecutive steps cannot cancel. Hence the corresponding path cannot immediately retrace an edge. If a reduced word represented the identity, then the corresponding path would begin and end at v, giving a closed path in T.

But a tree contains no nontrivial closed paths. Therefore such a reduced word cannot represent the identity. The only word which represents the identity is consequently the empty word.

Thus the chosen generators satisfy no nontrivial relations, and hence G is a free group.

20th Sep, 2026: ***Action of*** $SL(2,\mathbb{Z})$**.** We know that a group can be understood through the symmetries it produces. This suggests a natural question: what does it mean for $SL(2,\mathbb{Z})$ to act on a space?

Recall that

$$SL(2,\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a,b,c,d \in \mathbb{Z},\ ad - bc = 1 \right\}. \tag{27}$$

Since these are matrices, our natural candidate for an action is multiplication. If $v \in \mathbb{Z}^2$, we simply let

$$M \cdot v = Mv. \tag{28}$$

In other words, an element of $SL(2,\mathbb{Z})$ transforms a vector by the usual linear action of a matrix.

For our purposes, however, we are particularly interested in the primitive vectors

$$\mathbb{Z}^2_{\text{prim}} = \{(a,b) \in \mathbb{Z}^2 : \gcd(a,b) = 1\}. \tag{29}$$

The action restricts to this set because an element of $SL(2,\mathbb{Z})$ cannot turn a primitive vector into a non-primitive one. M^{-1} also has integer entries, so if Mv had a common

divisor, applying M^{-1} would give the same common divisor for the coordinates of v. Thus $SL(2,\mathbb{Z})$ acts naturally on a discrete collection of primitive vectors.

18th Sep, 2026: ***The Category of Sets.*** Let **Set** denote the category whose objects are sets and whose morphisms are functions. Then for $X, Y \in \mathbf{Set}$, $\mathrm{Hom}_{\mathbf{Set}}(X,Y)$ is the set of all functions $X \to Y$. We will talk about some universal structures in this category.

The *initial object* is $\varnothing$. And $|\mathrm{Hom}(\varnothing, X)| = 1$ for every X, since there is exactly one function with empty domain.

The *terminal objects* are the singleton sets. Let $*$ denote a singleton set, then $|\mathrm{Hom}(X,1)| = |X|$ for every X. Hence, **Set** has no zero object.

Next, let us consider *monomorphisms*.

A morphism $f : X \to Y$ is monic if

$$f \circ g = f \circ h \tag{30}$$

for all $g, h : Z \to X$ implies $g = h$. In **Set**, this happens when f is injective. Hence a subobject of X is represented by an injection $A \hookrightarrow X$, and, up to the categorical equivalence defining subobjects, they correspond to subsets $A \subseteq X$.

Dually, epimorphisms in **Set** are the surjective functions.

Then come *products and coproducts*. For sets X, Y, their product is the Cartesian product $X \times Y$ with projections

$$\pi_X : X \times Y \to X, \qquad \pi_Y : X \times Y \to Y. \tag{31}$$

Its universal property says that for every pair $f : Z \to X$ and $g : Z \to Y$, there exists a unique map $\langle f, g \rangle : Z \to X \times Y$ such that

$$\pi_X \langle f, g \rangle = f, \qquad \pi_Y \langle f, g \rangle = g. \tag{32}$$

The coproduct is the disjoint union $X \sqcup Y$, with canonical injections

$$i_X : X \to X \sqcup Y, \qquad i_Y : Y \to X \sqcup Y. \tag{33}$$

We then encounter *equalizers*. Given $f, g : X \to Y$, their equalizer is

$$E = \{x \in X \mid f(x) = g(x)\}, \tag{34}$$

together with the inclusion $E \hookrightarrow X$. Thus an equalizer extracts the locus on which two morphisms agree.

Its dual, the *coequalizer*, forms the quotient of Y by the smallest equivalence relation identifying $f(x) \sim g(x)$ for every $x \in X$.

From products and equalizers we obtain *pullbacks*, while from coproducts and coequalizers we obtain *pushouts*. In fact, **Set** possesses all small limits and all small colimits.

17th Sep, 2026: ***Isometric Group.*** For a metric space (X, d), a transformation preserving this structure is a map $f : X \to X$ satisfying $d(f(x), f(y)) = d(x, y)$ for all $x, y \in X$. Such a map is called an isometry. The set of all isometries of X is denoted by $\mathrm{Isom}(X)$.

The composition of two isometries is again an isometry, while the inverse of an isometry is also an isometry. Thus $\mathrm{Isom}(X)$ forms a group under composition.

16th Sep, 2026: ***Tiling, mathematically.*** We define the *barycentric subdivision* T' of a graph T to be the graph obtained by subdividing each edge. We then define a tile to be the subtree T_0 of T'. A *tiling* of T is then a collection of tiles satisfying the following:

(1) no two tiles share an edge, and
(2) the union of all tiles is the entire tree T'.

Given a group G which acts of the tree T, a *G-tiling* of T is a tiling of T satisfying an additional property:

(3) there exists a tile T_0 such that the set of all tiles is equal to $\{gT_0 : g \in G\}$.

15th Sep, 2026: ***Tiling of a Graph.*** Tiling a graph can be thought of as adding some 2-dimensional structure to an otherwise 1-dimensional object. A graph gives vertices and edges, but certain collections of edges can form meaningful configurations, such as triangles. These configurations can be regarded as pieces, or tiles, attached to the graph.

The point of doing this is that the tiles encode relations between paths. For example, a triangle gives a closed loop, and regarding it as a tile allows that loop to be treated as a basic relation.

Thus, tiling gives a way of passing from the graph itself to a richer geometric object, where the way the pieces fit together contains information about the global structure of the graph. For a graph, the underlying object is the graph itself, and a tiling specifies a collection of subgraphs that serve as the tiles. These tiles may intersect along common vertices or edges, and their union recovers the graph.

14th Sep, 2026: ***Farey Tree.*** For a Farey tree, we begin by looking at the triangular regions that appear in the Farey graph. Rather than leaving these regions as empty spaces, we can fill them in and think of each triangle as a solid planar piece. In other words, we keep the vertices and edges of the Farey graph, but now regard the triangular regions bounded by those edges as part of the object itself.

Once we do this for every triangular region, we obtain a 2-dimensional simplicial complex. This is what we call the Farey complex. From here, we can get the Farey tree by looking at the Farey complex in a slightly different way. Instead of putting our attention on the vertices of the triangles, we can place a new vertex inside each triangular face. We then connect two new vertices whenever the corresponding triangles share an edge.

In this way, the triangles of the Farey complex become the vertices of a new graph, and the way in which the triangles meet tells us which vertices to connect. The resulting graph is the Farey tree.

13th Sep, 2026: ***Drinfeld Center.*** Let $\mathscr{C}$ be a monoidal category.

The delooping $B\mathscr{C}$ is the category with a single object $*$ whose endomorphisms are given by the objects of $\mathscr{C}$:

$$\mathrm{End}_{B\mathscr{C}}(*) \simeq \mathscr{C}. \tag{35}$$

In this picture, composition of morphisms in $B\mathscr{C}$ corresponds to the tensor product in $\mathscr{C}$. Thus, delooping shifts the monoidal structure up by one categorical level: objects of $\mathscr{C}$ are viewed as morphisms of the single object of $B\mathscr{C}$.

Now, an object $X \in \mathscr{C}$ should be *central* if it can be moved past every other object $Y \in \mathscr{C}$, in a way that is compatible with the monoidal structure.

An object of the Drinfeld center $Z(\mathscr{C})$ is a pair (X, γ), where $X \in \mathscr{C}$ and γ is a natural family of isomorphisms

$$\gamma_Y : X \otimes Y \xrightarrow{\sim} Y \otimes X, \tag{36}$$

one for every $Y \in \mathscr{C}$. The family γ is called a half-braiding.

The isomorphisms γ_Y cannot be chosen independently. They must satisfy a coherence condition expressing compatibility with tensor products. In the strict monoidal case,

$$\gamma_{Y\otimes Z} = (\mathrm{id}_Y \otimes \gamma_Z) \circ (\gamma_Y \otimes \mathrm{id}_Z). \tag{37}$$

Therefore, the Drinfeld center can be thought of as the category of objects of $\mathscr{C}$ which are coherently commutative with every other object.

We had also discussed the center of a category here 2. Something to note here is that for an ordinary category $\mathscr{C}$, its center consists of natural endomorphisms of the identity functor, so an element of the center assigns to every object X a morphism $X \to X$ which commutes naturally with every morphism in $\mathscr{C}$. The Drinfeld center is essentially a categorified version of this idea. If $\mathscr{C}$ is monoidal, we can deloop it to obtain $B\mathscr{C}$, where the single object has $\mathscr{C}$ as its endomorphisms. Thus, centrality in $B\mathscr{C}$ means that an object $X \in \mathscr{C}$ should commute with every other object Y, but now this commutation is expressed by coherent isomorphisms $X \otimes Y \simeq Y \otimes X$. These coherent isomorphisms are precisely the half-braiding defining an object of $Z(\mathscr{C})$.

11th Sep, 2026: ***Affine Scheme and Scheme.*** Let R be a commutative ring. The *affine scheme* associated to R is the locally ringed space

$$\mathrm{Spec}(A) = \big(|\,\mathrm{Spec}(A)|, \mathcal{O}_{\mathrm{Spec}(A)}\big), \tag{38}$$

where

$$|\,\mathrm{Spec}(A)| = \{\mathfrak{p} \subseteq A : \mathfrak{p} \text{ is a prime ideal}\} \tag{39}$$

is equipped with the Zariski topology, whose closed subsets are

$$V(I) = \{\mathfrak{p} \in \mathrm{Spec}(A) : I \subseteq \mathfrak{p}\}, \tag{40}$$

and $\mathcal{O}_{\mathrm{Spec}(A)}$ is the structure sheaf defined on the basic open sets

$$D(f) = \{\mathfrak{p} \in \mathrm{Spec}(A) : f \notin \mathfrak{p}\} \tag{41}$$

by $\mathcal{O}_{\mathrm{Spec}(A)}(D(f)) = A_f$.

A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ such that for every $x \in X$, there exists an open neighborhood $U \subseteq X$ and a commutative ring A such that

$$(U, \mathcal{O}_X|_U) \cong (\mathrm{Spec}(A), \mathcal{O}_{\mathrm{Spec}(A)}) \tag{42}$$

as locally ringed spaces.

For $f \in A$, let

$$D(f) = \{\mathfrak{p} \in \mathrm{Spec}(A) : f \notin \mathfrak{p}\}. \tag{43}$$

The sets $D(f)$ form a basis for the Zariski topology. We define

$$\mathcal{O}_{\mathrm{Spec}(A)}(D(f)) := A_f, \tag{44}$$

where A_f is the localization of A at $\{1, f, f^2, \ldots\}$.

10th Sep, 2026: ***Symmetric Monoidal Categories.*** Let $\mathscr{C}$ be a monoidal category. A symmetric monoidal structure on $\mathscr{C}$ is a natural isomorphism

$$\sigma_{X,Y} : X \otimes Y \xrightarrow{\sim} Y \otimes X \tag{45}$$

such that

$$\sigma_{Y,X} \circ \sigma_{X,Y} = \mathrm{id}_{X \otimes Y} \tag{46}$$

for all $X, Y \in \mathscr{C}$, and such that the following hexagon diagram commutes:

$$\begin{array}{ccc} (X \otimes Y) \otimes Z & \xrightarrow{\alpha_{X,Y,Z}} & X \otimes (Y \otimes Z) \\ {\scriptstyle \sigma_{X,Y} \otimes \mathrm{id}_Z}\downarrow & & \downarrow{\scriptstyle \mathrm{id}_X \otimes \sigma_{Y,Z}} \\ (Y \otimes X) \otimes Z & & X \otimes (Z \otimes Y) \\ {\scriptstyle \alpha_{Y,X,Z}}\downarrow & & \uparrow{\scriptstyle \alpha_{Y,Z,X}} \\ Y \otimes (X \otimes Z) & \xrightarrow[\sigma_{X,Y \otimes Z}]{} & (Y \otimes Z) \otimes X \end{array} \tag{47}$$

A monoidal category equipped with such a symmetric monoidal structure is called a *Symmetric monoidal category.*

9th Sep, 2026: ***Monoidal Categories.*** A monoidal category is, roughly, a category in which we have some notion of putting objects together. This is encoded by a tensor product

$$\otimes : \mathscr{C} \times \mathscr{C} \to \mathscr{C}. \tag{48}$$

There is also a distinguished object $\mathbf{1}$, which takes everything to itself.

The tensor product behaves associatively, at least up to a specified isomorphism:

$$(X \otimes Y) \otimes Z \cong X \otimes (Y \otimes Z), \tag{49}$$

and similarly tensoring with $\mathbf{1}$ does not change an object, up to isomorphism.

However, there is no reason for the tensor product to be commutative. In general, there need not be any natural isomorphism

$$X \otimes Y \cong Y \otimes X. \tag{50}$$

So a monoidal category gives us a setting where we can combine things, but does not automatically tell us that the order in which we combine them is irrelevant.

This is precisely when we start looking for the notion of the center of a category discussed here(2).

8th Sep, 2026: ***Hilbert Nullstellensatz(s).*** We list the statements of the weak form and the strong form of Hilbert Nullstellansatz.

Weak Hilbert Nullstellensatz. Let k be an algebraically closed field. Then every maximal ideal $\mathfrak{m}$ of $k[x_1, \ldots, x_n]$ is of the form

$$\mathfrak{m} = (x_1 - a_1, \ldots, x_n - a_n) \tag{51}$$

for some $a_1, \ldots, a_n \in k$.

Equivalently, for every proper ideal $\mathfrak{I} \subset k[x_1, \ldots, x_n]$, $V(\mathfrak{I}) \neq \varnothing$, where

$$V(\mathfrak{I}) = \{(a_1, \ldots, a_n) \in k^n : f(a_1, \ldots, a_n) = 0 \ \forall f \in \mathfrak{I}\}. \tag{52}$$

Strong Hilbert Nullstellensatz. Let k be an algebraically closed field and let $\mathfrak{I} \subseteq k[x_1, \ldots, x_n]$ be an ideal. Then $\mathfrak{I}(V(\mathfrak{I})) = \sqrt{\mathfrak{I}}$, where

$$\mathfrak{I}(V(\mathfrak{I})) = \{f \in k[x_1, \ldots, x_n] : f(a) = 0 \ \forall a \in V(\mathfrak{I})\} \tag{53}$$

and

$$\sqrt{\mathfrak{I}} = \{f \in k[x_1, \ldots, x_n] : f^m \in \mathfrak{I} \ \forall m \geq 1\}. \tag{54}$$

Equivalent polynomial formulation. Let k be algebraically closed, $\mathfrak{I} \subseteq k[x_1, \ldots, x_n]$ an ideal, and $f \in k[x_1, \ldots, x_n]$. Then $f(a) = 0$ for every $a \in V(\mathfrak{I})$ iff $f^m \in \mathfrak{I}$ for some $m \geq 1$.

7th Sep, 2026: ***Understanding Tannaka.*** Tannaka duality is, roughly, the idea that we can understand an algebraic object by looking at its representations. Instead of studying a group directly, we study the category of objects on which the group acts, together with the way these representations can be forgotten back to ordinary vector spaces.

Under suitable conditions, this representation-theoretic data contains enough information to reconstruct the original group itself:

$$\text{algebraic object} \quad \longleftrightarrow \quad \text{its representations.}$$

This viewpoint is one of the places where category theory, representation theory, and algebraic geometry naturally meet.

6th Sep, 2026: ***The Farey Graph.*** The Farey graph is a way of putting a graph structure on the set of rational numbers.

Suppose that $\frac{p}{q}$ and $\frac{r}{s}$ are reduced fractions. We join them by an edge when $|ps - qr| = 1$. Thus the edges of the graph are determined by the determinant

$$\det \begin{pmatrix} p & r \\ q & s \end{pmatrix} = ps - qr. \tag{55}$$

The Farey complex is the simplicial complex obtained from the Farey graph by filling in every triangle whose three vertices are pairwise adjacent. Its vertices are $\mathbb{Q} \cup \{\infty\}$, and two vertices $\frac{p}{q}$ and $\frac{r}{s}$ are joined whenever

$$|ps - qr| = 1. \tag{56}$$

A 2-simplex is then a triple

$$\frac{p}{q}, \qquad \frac{r}{s}, \qquad \frac{p+r}{q+s}, \tag{57}$$

where $\frac{p}{q}$ and $\frac{r}{s}$ are adjacent. Thus the Farey complex can be viewed as the triangulation obtained by filling the Farey graph with all such Farey triangles.

5th Sep, 2026: ***Center of a category.*** Let $\mathscr{C}$ be a category. Now, we look for an automorphism of every object that is compatible with every morphism in the category.

Suppose that for each object X we choose an isomorphism

$$\eta_X : X \longrightarrow X. \tag{58}$$

We want these isomorphisms to behave coherently with respect to the morphisms of $\mathscr{C}$. Thus, for every morphism $f : X \to Y$, we require

$$f \circ \eta_X = \eta_Y \circ f. \tag{59}$$

In particular, the collection $\{\eta_X\}_{X \in \mathscr{C}}$ commutes, in a categorical sense, with every morphism.

Note that this condition is exactly the condition for η to be a natural transformation from the identity functor to itself. Moreover, since each η_X is an isomorphism, η is a natural automorphism of the identity functor. We therefore define the center of $\mathscr{C}$ by

$$Z(\mathscr{C}) = \mathrm{Aut}(\mathrm{Id}_{\mathscr{C}}). \tag{60}$$

The analogy with groups can therefore be summarized as follows. For a group, the center consists of elements that commute with every element:

$$Z(G) = \{g \in G : gh = hg \text{ for all } h \in G\}. \tag{61}$$

For a category, the center consists of natural automorphisms of the identity functor, whose components commute with every morphism:

$$Z(\mathscr{C}) = \mathrm{Aut}(\mathrm{Id}_{\mathscr{C}}). \tag{62}$$

4th Sep, 2026: ***More on germs and rational maps.*** A germ is local at a particular point, whereas a rational map records what happens on a dense open subset of the variety. In this sense, one can think of a rational map as a kind of "generic germ."

For an irreducible variety X, this analogy becomes particularly precise through the generic point. Let $\mathcal{O}_X$ denote the structure sheaf of X, and let η be the generic point of X. Then the stalk of the structure sheaf at η is precisely the function field:

$$\mathcal{O}_{X,\eta} = K(X). \tag{63}$$

Thus, the function field can be viewed as the ring of functions defined near the generic point. This gives a useful parallel: an ordinary stalk records the behaviour of functions near a fixed point, while the stalk at the generic point records their generic behaviour over the whole variety.

The underlying principle is therefore the same. A germ identifies functions which become equal after restricting to a sufficiently small neighbourhood, while a rational map identifies functions which become equal after restricting to a sufficiently large, nonempty open subset.

3rd Sep, 2026: ***On rational maps.*** A rational map is, roughly speaking, a map which is allowed to be undefined on a small part of its domain.

Let X and Y be algebraic varieties. A rational map from X to Y is a regular map defined on some nonempty Zariski-open subset of X. Thus, we write

$$f : X \dashrightarrow Y \tag{64}$$

to indicate that there is some nonempty open subset $U \subseteq X$ for which

$$f : U \longrightarrow Y \tag{65}$$

is a regular map. However, the particular choice of U is not part of the data we ultimately care about. If

$$f : U \longrightarrow Y \qquad \text{and} \qquad g : V \longrightarrow Y \tag{66}$$

are regular maps, then we regard them as defining the same rational map if they agree on some nonempty open subset of $U \cap V$. Thus, a rational map is really an equivalence class of regular maps which agree wherever they are both defined, after possibly restricting to a smaller open set.

For example, consider the expression $f(x) = \frac{x^2-1}{x-1}$. This does not define a regular function on all of $\mathbb{A}^1$, since it is undefined at $x = 1$. On the other hand, on the open subset $U = \mathbb{A}^1 \setminus 1$, we have

$$\frac{x^2-1}{x-1} = x + 1.$$

Consequently, the rational functions represented by $\frac{x^2-1}{x-1}$ and $x+1$ are the same. Rational functions do not care about what happens on a proper closed subset, but about their behaviour on a nonempty open subset.

This has a *close analogy* with the sheaf-theoretic notion of a germ. If $\mathcal{F}$ is a sheaf on a space X, and $s \in \mathcal{F}(U)$, then the germ of s at a point $x \in U$ is an element $[s]_x \in \mathcal{F}_x$, where the stalk is given by

$$\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U). \tag{67}$$

The idea is that two sections determine the same germ at x if they agree on some sufficiently small neighbourhood of x. A germ therefore forgets the global behaviour of a section and remembers only its behaviour arbitrarily close to one particular point.

2nd Sep, 2026: *On regular morphisms.* A *regular morphism* between algebraic varieties is, roughly, a map that is locally described by regular functions. On an affine variety, these are functions that can locally be written as quotients of polynomials, with the denominator non-zero at the point in question. Thus, regular morphisms provide the natural notion of a map that respects the algebraic structure of varieties.

For affine varieties $X \subseteq \mathbb{A}^n$ and $Y \subseteq \mathbb{A}^m$, a regular morphism $f : X \to Y$ induces a k-algebra homomorphism

$$f^* : k[Y] \longrightarrow k[X], \qquad g \mapsto g \circ f. \tag{68}$$

Conversely, every k-algebra homomorphism $k[Y] \to k[X]$ gives a regular morphism $X \to Y$.

1st Sep, 2026: ***Subsheaves and Subpresehaves.*** An interesting thing to note in algebraic geometry would be to note that the catgeory of subsheaves will form a full subcategory of the catgeory of subpresheaves. The analogue of this statement for a variety X translates to a concrete property about subobjects of a fixed sheaf (like the structure sheaf of regular functions $\mathcal{O}_X$), i.e., if we have two subsheaves, they are automatically subpresheaves. This statement means that any map between them that preserves their values on open sets is automatically a valid geometric map between them as subsheaves.

31st Aug, 2026: ***The action of Free groups.*** An interesting thing to note about free groups and their actions is this theorem which states that *if a group G acts freely on a tree then G is a free group* and its converse. Reducing the theorem to the statment that *G being free is equivalent to the condition that G acts freely on a tree.*

30th Aug, 2026. A quadratic form on $\mathbb{R}^n$ is an expression of the form $Q(x) = x^T Ax$, where A may always be taken to be symmetric.

If the eigenvalues of A are $\lambda_1, \ldots, \lambda_n$, then, after choosing an orthonormal basis of eigenvectors, the quadratic form takes the simpler form $Q(x) = \lambda_1 x_1^2 + \cdots + \lambda_n x_n^2$. Thus, the signs of the eigenvalues determine whether Q is positive definite, negative definite, or indefinite.

The Rayleigh quotient $R_A(x) = \frac{x^T Ax}{x^T x}$, for $x \neq 0$, gives a normalized measure of the quadratic form in the direction of x. In fact, when A is symmetric, $\lambda_{\min} \leq R_A(x) \leq \lambda_{\max}$, with equality occurring precisely in the corresponding eigendirections.

29th Aug, 2026. Some definitions useful in algebraic geometry, when talking of varieties, have been discussed below.

Let $K \subseteq L$ be a field extension, and let $\alpha \in L$. We say that α is *algebraic over K* if there exists a nonzero polynomial $f(x) \in K[x]$ such that $f(\alpha) = 0$.
If no such nonzero polynomial exists, then α is said to be *transcendental over K*.

Let $K \subseteq L$ be a field extension. The extension L/K is called *algebraic* if every element of L is algebraic over K. That is, for every $\alpha \in L$, $\exists\, 0 \neq f(x) \in K[x]$ such that $f(\alpha) = 0$.

The extension L/K is called *transcendental* if it is not algebraic. Equivalently, if there exists some $\alpha \in L$ that is transcendental over K.

Let $K \subseteq L$ be a field extension, and let $S \subseteq L$. The set S is said to be *algebraically independent over K* if, for every $n \geq 1$, every choice of pairwise distinct elements $\alpha_1, \ldots, \alpha_n \in S$, and every polynomial $f \in K[x_1, \ldots, x_n]$, the implication

$$f(\alpha_1, \ldots, \alpha_n) = 0 \quad \Longrightarrow \quad f = 0 \tag{69}$$

holds.
If S is not algebraically independent over K, then S is said to be *algebraically dependent over K*.

Let $K \subseteq L$ be a field extension. A subset $B \subseteq L$ is called a *transcendence basis of L over K* if:

(1) B is algebraically independent over K; and
(2) L is algebraic over $K(B)$.

Let $K \subseteq L$ be a field extension, and let B be a transcendence basis of L over K. The *transcendence degree of L over K* is defined by

$$\operatorname{trans}_K L = |B|. \tag{70}$$

Equivalently, it is the cardinality of any maximal algebraically independent subset of L over K.

28th Aug, 2026. Let $S^1 = \{z \in \mathbb{C} : |z| = 1\}$, with basepoint 1. We prove that

$$\pi_1(S^1, 1) \cong \mathbb{Z}. \tag{71}$$

For each $n \in \mathbb{Z}$, define $\gamma_n : [0,1] \to S^1$, $\gamma_n(t) = e^{2\pi i n t}$.

Since

$$\gamma_n(0) = 1 = \gamma_n(1), \tag{72}$$

γ_n is a loop based at 1.

Define

$$\Phi : \mathbb{Z} \to \pi_1(S^1, 1), \qquad \Phi(n) = [\gamma_n]. \tag{73}$$

Let

$$p : \mathbb{R} \to S^1, p(x) = e^{2\pi i x}. \tag{74}$$

Every loop $\alpha : [0,1] \to S^1$ based at 1 has a unique lift

$$\widetilde{\alpha} : [0,1] \to \mathbb{R} \tag{75}$$

such that $\widetilde{\alpha}(0) = 0$.

Since $\alpha(1) = 1$, $e^{2\pi i \widetilde{\alpha}(1)} = 1$, and hence $\widetilde{\alpha}(1) \in \mathbb{Z}$.

Define

$$W : \pi_1(S^1, 1) \to \mathbb{Z}, \qquad W([\alpha]) = \widetilde{\alpha}(1). \tag{76}$$

This is well-defined since based-homotopic loops have lifts with the same endpoint.

For γ_n, the lift satisfying $\widetilde{\gamma_n}(0) = 0$ is $\widetilde{\gamma_n}(t) = nt$.

Therefore $W(\Phi(n)) = W([\gamma_n]) = \widetilde{\gamma_n}(1) = n$.

Hence $W \circ \Phi = \mathrm{id}_{\mathbb{Z}}$.

Conversely, let $[\alpha] \in \pi_1(S^1, 1)$, and suppose $W([\alpha]) = n$.

Then $\widetilde{\alpha}$ is a path in $\mathbb{R}$ from 0 to n.

Define

$$H : [0,1] \times [0,1] \to \mathbb{R} \text{ by } H(s,t) = (1-s)\widetilde{\alpha}(t) + snt. \tag{77}$$

Then $H(0,t) = \widetilde{\alpha}(t)$, $H(1,t) = nt$, and, for every $s \in [0,1]$, $H(s,0) = 0$, $H(s,1) = n$.

Thus H is a homotopy relative to the endpoints between $\widetilde{\alpha}$ and the path $t \mapsto nt$.

Applying p, we obtain a based homotopy between α and γ_n. Hence $[\alpha] = [\gamma_n] = \Phi(n)$.

Therefore $\Phi \circ W = \mathrm{id}_{\pi_1(S^1,1)}$.

Consequently, Φ and W are inverse isomorphisms, and hence

$$\pi_1(S^1, 1) \cong \mathbb{Z}.$$

□

27th Aug, 2026. Reporting a work in progress for this day and the coming day.

I have been trying to list out all the elements of the automorphism group of a cube explicitly, and here is my attempt at that-link.

26th Aug, 2026. Divisibility and ideals are closely related. In $\mathbb{Z}$, the statement $a \mid b$ means that there exists some $k \in \mathbb{Z}$ such that $b = ak$. Equivalently, $b \in \langle a \rangle$, where $\langle a \rangle = \{ak : k \in \mathbb{Z}\}$ is the principal ideal generated by a. Thus, divisibility by a may be understood as membership in the principal ideal generated by a.

This relationship also explains the connection between divisibility and inclusion of ideals. Suppose that $a \mid b$. Then $b = ak$ for some $k \in \mathbb{Z}$. Every multiple of b is therefore also a multiple of a, and hence $\langle b \rangle \subseteq \langle a \rangle$. Thus divisibility gives an inclusion of principal ideals, although the direction of inclusion is reversed: $a \mid b \Longrightarrow \langle b \rangle \subseteq \langle a \rangle$.

This point of view extends naturally to an arbitrary commutative ring R. For $a \in R$, the principal ideal generated by a is $\langle a \rangle = \{ra : r \in R\}$. Hence, for $a, b \in R$, $a \mid b \Longleftrightarrow b \in \langle a \rangle$. The notion of a principal ideal therefore retains the familiar idea of collecting together all multiples of a single element.

However, ideals are more general than principal ideals. An ideal need not be generated by a single element. Given elements $a_1, \ldots, a_n \in R$, the ideal they generate is $(a_1, \ldots, a_n) = \{r_1a_1 + \cdots + r_na_n : r_1, \ldots, r_n \in R\}$. Thus, while principal ideals arise naturally from divisibility by a single element, general ideals allow several elements to generate a common algebraic object.

In this way, ideals may be viewed as extending the idea of divisibility.

25th Aug, 2026. Let $F_n = \langle x_1, \ldots, x_n \rangle$ be the free group on n generators. Since every element of F_n has a unique reduced word, we can describe conjugacy classes in terms of these words.

Two elements $g, h \in F_n$ are conjugate if there exists some $w \in F_n$ such that $h = wgw^{-1}$. Thus, a conjugacy class is $[g] = \{wgw^{-1} : w \in F_n\}$.

Since F_n is a free group, every element has a unique reduced word. A reduced word

$$u = x_{i_1}^{\epsilon_1} \cdots x_{i_k}^{\epsilon_k}$$

is called *cyclically reduced* if its first letter is not the inverse of its last letter. Every element of F_n is conjugate to a cyclically reduced word. Indeed, if the first and last letters of a reduced word cancel when the word is considered cyclically, we can conjugate away these two letters. Repeating this process eventually gives a cyclically reduced word.

For example, $x_1x_2x_1^{-1}x_2^{-1}$ is already cyclically reduced, whereas $x_1x_2x_1^{-1}$ is not, since its first and last letters are inverses. In this case, $x_1x_2x_1^{-1} = x_1(x_2)x_1^{-1}$, so it is conjugate to the shorter cyclically reduced word x_2.

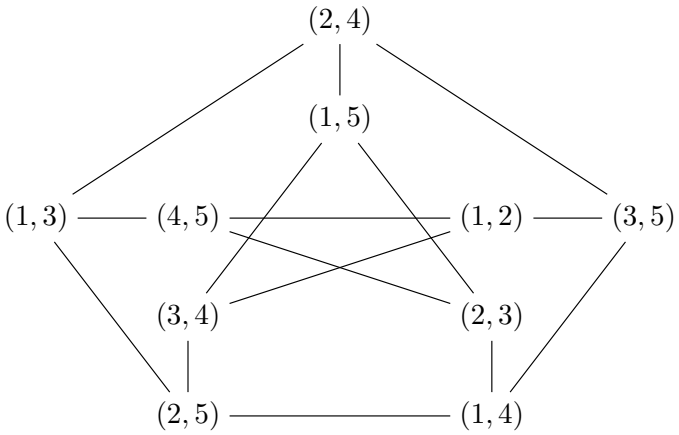
Now suppose that u and v are cyclically reduced. Then u and v are conjugate iff v is a cyclic permutation of u. For example, $x_1x_2x_1^{-1}x_2^{-1}$ is conjugate to $x_2x_1^{-1}x_2^{-1}x_1$, because the latter is obtained by cyclically permuting the former.

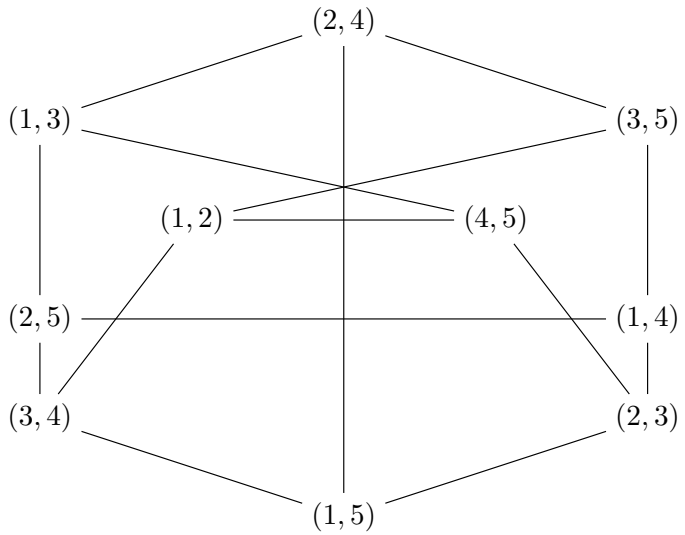
However, that this does not mean that a conjugacy class is finite. The cyclically reduced word only gives us a convenient representative of the conjugacy class. In fact, for $n \geq 2$, every nontrivial conjugacy class in F_n is infinite. For example, in $F_2 = \langle a, b \rangle$, the elements $a, bab^{-1}, b^2ab^{-2}, b^3ab^{-3}, \ldots$ are all conjugate to a, and they are all distinct.

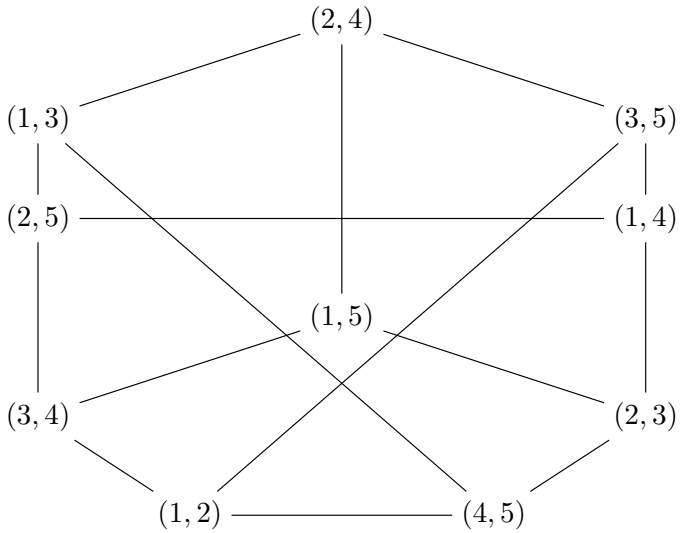
Hence, apart from the identity, the conjugacy classes of F_n can be described by nonempty cyclically reduced words modulo cyclic permutation, even though each such conjugacy class contains infinitely many elements.

24th Aug, 2026. For the sake of *fun*, I tried showing the isomorphisms in Petersen graphs via quiver diagrams. Essentially, all the following graphs are the same up to rearrangement

of vertices.


(2, 4)
(1, 5)
(1, 3)
(4, 5)
(1, 2)
(3, 5)
(3, 4)
(2, 3)
(2, 5)
(1, 4)


(2, 4)
(1, 3)
(3, 5)
(1, 2)
(4, 5)
(2, 5)
(1, 4)
(3, 4)
(2, 3)
(1, 5)


(2, 4)
(1, 3)
(3, 5)
(2, 5)
(1, 4)
(1, 5)
(3, 4)
(2, 3)
(1, 2)
(4, 5)

23rd Aug, 2026. In Fourier theory, the basic objects are characters of a group: homomorphisms from a group G into the circle group $S^1 = \{z \in \mathbb{C} : |z| = 1\}$. Thus a character is a map

$$\chi : G \to S^1 \tag{78}$$

satisfying $\chi(gh) = \chi(g)\chi(h)$. The group operation is translated into multiplication of complex numbers, allowing algebraic information about G to be studied through functions.

This naturally leads to the question of how much information these characters contain about the group. For abelian groups, characters are particularly well behaved: they form the appropriate analogue of the Fourier basis, and functions on the group can be decomposed in terms of them. This is the beginning of character theory, where we study these homomorphisms and the ways in which they reflect the structure of the underlying group. For non-abelian groups, the same philosophy leads us to higher-dimensional representations and, eventually, to the more general notion of a character.

22nd Aug, 2026. Let R be a commutative ring and $I \subset R$ an ideal. An ideal is, in a sense, the basic algebraic object through which we study the structure of a ring, but not all ideals behave equally well with respect to multiplication. The distinction becomes apparent when we consider a product $ab \in I$. For an arbitrary ideal, there is no reason for either a or b to belong to I. However, a prime ideal imposes the strongest possible condition: an ideal $\mathfrak{p} \subset R$ is prime if

$$ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}. \tag{79}$$

Equivalently, $R/\mathfrak{p}$ is an integral domain. Thus, prime ideals are those ideals for which the quotient has no non-zero zero divisors.

Primary ideals weaken this condition slightly. A proper ideal $Q \subset R$ is called primary if

$$ab \in Q \implies a \in Q \text{ or } b^n \in Q \tag{80}$$

for some $n \geq 1$. Hence a primary ideal allows a product to lie in Q even when neither factor itself lies in Q, provided that one of the factors becomes an element of Q after taking a sufficiently large power.

Every prime ideal is therefore primary, but the converse need not hold. For instance,

$$\langle 4 \rangle \subseteq \mathbb{Z}$$

is not prime, since $2 \cdot 2 \in \langle 4 \rangle$ while $2 \notin \langle 4 \rangle$, but it is primary since whenever $ab \in \langle 4 \rangle$, one of a or b is divisible by 2, and hence a *suitable* power of that element is divisible by 4.

Hence, an ideal Q is primary precisely when every zero divisor in R/Q is nilpotent, whereas a prime ideal gives a quotient which is an integral domain. This provides the useful correspondence

$$Q \text{ primary} \quad \iff \quad \text{every zero divisor of } R/Q \text{ is nilpotent.} \tag{81}$$

21st Aug, 2026. A topological space X is called *irreducible* if it is nonempty and cannot be written as the union of two proper closed subsets, i.e. whenever $X = Z_1 \cup Z_2$ with Z_1, Z_2 closed, then $X = Z_1$ or $X = Z_2$.

Equivalently, every two nonempty open subsets of X have nonempty intersection. Thus, irreducibility is stronger than connectedness, since every irreducible space is connected, but the converse need not hold.

For a topological space X, its dimension is defined as the supremum of the integers n for which there exists a chain of irreducible closed subsets

$$X_0 \subset X_1 \subset \cdots \subset X_n. \tag{82}$$

Thus, $\dim X$ measures the maximum length of a strict chain of irreducible closed subsets. In particular, a point has dimension 0, an irreducible curve has dimension 1, and $\mathbb{A}^n$ has dimension n.

For an affine variety $X = V(I) \subseteq \mathbb{A}^n_k$ with coordinate ring $k[X] = k[x_1, \ldots, x_n]/I$, its dimension is defined by

$$\dim X = \dim k[X]. \tag{83}$$

Equivalently, $\dim X$ is the supremum of the lengths of chains of irreducible closed subsets (as discussed above)

$$X_0 \subset X_1 \subset \cdots \subset X_n. \tag{84}$$

Algebraically, this corresponds to chains of prime ideals in $k[X]$.

For a ring A, the *Krull dimension* $\dim A$ is the supremum of the lengths of chains of prime ideals

$$\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_n. \tag{85}$$

The appearance of prime ideals comes from the correspondence between prime ideals of A and irreducible closed subsets of $\operatorname{Spec} A$, where $\mathfrak{p}$ corresponds to $V(\mathfrak{p})$. Thus, the Krull dimension of A agrees with the dimension of its spectrum.

20th Aug, 2026. We describe, in brief, about the *Pontryagin duality*, as an attempt to understand something that comes after.

(1) A *topological group* G is a topological space and a group, such that the group operation is defined as

$$\cdot : G \times G \to G \text{ such that } (x, y) \mapsto xy \tag{86}$$

and the inverse map

$$^{-1}\colon G \to G \text{ such that } x \mapsto x^{-1} \tag{87}$$

are continuous.

We view $G \times G$ as a topological space with the product topology.

(2) Let X be a topological space. X is said to be *locally compact* if every point $x \in X$ has a compact neighbourhood.

(3) A *Hausdorff space* in a topological space where distinct points have disjoint neighbourhoods. We also call this a separated space.
(4) A *locally compact group* is a topological group for which the underlying topology is locally compact and Hausdorff.
(5) For a locally compact abelian group G, the group of continuous homomorphisms $\mathrm{Hom}(G, S^1)$ is locally compact.
(6) Pontryagin Duality says that the following functor induces an equivalence of categories.

$$\mathrm{Cat}(\text{Locally Compact Abelian Groups})^{op} \to \mathrm{Cat}(\text{Locally Compact Abelian Groups}) \tag{88}$$

(7) An *equivalence of categories* consists of the following:
   - two catgeories, say $\mathscr{C}$ and $\mathscr{D}$,
   - two functors, say $F : \mathscr{C} \to \mathscr{D}$ and $G : \mathscr{D} \to \mathscr{C}$, and
   - two natural isomorphisms, say $\eta : FG \to I_{\mathscr{D}}$ and $\varepsilon : I_{\mathscr{C}} \to GF$, where I_{cat} denotes the identity functor omn the respective category.

   Equivalently, a functor is said to establish an equivalence of catgeories if it is full, faithful and essentially surjective.

We will perhaps discuss more about this in the upcoming days, as I try to build some substantial understanding fo the subject.

19th Aug, 2026. For a ring R, R is Noetherian $\iff$ $I_1 \subseteq I_2 \subseteq \cdots$ stabilizes, i.e. R satisfies the ACC on ideals.

For $X = \operatorname{Spec} R$, ideals correspond contravariantly to closed sets:

$$I \subseteq J \quad \Longrightarrow \quad V(J) \subseteq V(I). \tag{89}$$

Thus

$$I_1 \subseteq I_2 \subseteq \cdots \quad \Longrightarrow \quad V(I_1) \supseteq V(I_2) \supseteq \cdots . \tag{90}$$

The order reverses because Spec is contravariant. We then arrive at the following definition.

A topological space X is called *Noetherian* if every descending chain of closed subsets $X_1 \supseteq X_2 \supseteq X_3 \supseteq \cdots$ stabilizes, that is, there exists $n \in \mathbb{N}$ such that $X_n = X_{n+1} = X_{n+2} = \cdots$.

Dually, a topological space X is called *Artinian* if every ascending chain of closed subsets $X_1 \subseteq X_2 \subseteq X_3 \subseteq \cdots$ stabilizes, that is, there exists $n \in \mathbb{N}$ such that $X_n = X_{n+1} = X_{n+2} = \cdots$.

Now, noetherian spaces occur naturally in algebraic geometry, at least, most of the notions are defined on noetherian topological spaces. It is useful because it gives finiteness, such that descending chains of closed sets terminates, every closed subset has finitely many irreducible components, and every open subset is quasi-compact[1].

[1]This is the same as compactness in metric spaces.

Naturally, a question arises to observe the theory of Artinian topological spaces and if it admits any stronger results than noetherian topological spaces.

18th Aug, 2026. An adjunction between categories $\mathscr{C}$ and $\mathscr{D}$ consists of functors $F : \mathscr{C} \to \mathscr{D}$ and $G : \mathscr{D} \to \mathscr{C}$ together with a natural bijection

$$\mathrm{Hom}_{\mathscr{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathscr{C}}(X, G(Y)). \tag{91}$$

We write $F \dashv G$, calling F the left adjoint and G the right adjoint.

The adjunction expresses a universal relationship between the two functors: morphisms out of $F(X)$ correspond naturally to morphisms into $G(Y)$. This correspondence is encoded by the unit and counit of the adjunction and often allows constructions in one category to be translated into another.

A fundamental consequence is that left adjoints preserve colimits, while right adjoints preserve limits.

If $F \dashv G$ and $D : I \to \mathscr{C}$ has a colimit $\operatorname{colim} D$, then

$$F(\mathrm{colim}_{i\in I}\, D(i)) \cong \mathrm{colim}_{i\in I}\, F(D(i)) \tag{92}$$

The reason being that, for every $Y \in \mathscr{D}$,

$$\begin{aligned} \mathrm{Hom}_{\mathscr{D}}\left(F\left(\mathrm{colim}_i\, \mathscr{D}(i)\right), Y\right) &\cong \mathrm{Hom}_{\mathscr{C}}\left(\mathrm{colim}_i\, \mathscr{D}(i), G(Y)\right) \\ &\cong \lim_i \mathrm{Hom}_{\mathscr{C}}\left(\mathscr{D}(i), G(Y)\right) \\ &\cong \lim_i \mathrm{Hom}_{\mathscr{D}}\left(F(\mathscr{D}(i)), Y\right) \\ &\cong \mathrm{Hom}_{\mathscr{D}}\left(\mathrm{colim}_i\, F(\mathscr{D}(i)), Y\right). \end{aligned} \tag{93}$$

The last equality precisely gives us the universal property of the colimit.

17th Aug, 2026. Some mathematical ideas which I revisited on this day are:

(1) A subgroup generated by a set S is the smallest subgroup containing S. Equivalently, it consists of all finite products of elements of S and their inverses. To prove that S generates a group G, one must show that every element of G can be written as such a product. When proving equality of a generated subgroup with G, both inclusions should be established explicitly. Usually, the inclusion $\langle S\rangle \subseteq G$ is immediate, while $G \subseteq \langle S\rangle$ contains the main argument.

(2) In $\mathbb{Z}/n\mathbb{Z}$, the subgroup generated by $\overline{m}$ is

$$\langle \overline{m}\rangle = \{k\overline{m} : k \in \mathbb{Z}\}.$$

If $\gcd(m, n) = 1$, Bézout's identity gives integers a, b satisfying $am + bn = 1$. Passing to the quotient gives $\overline{a}\,\overline{m} = \overline{1}$. Hence $\overline{1} \in \langle \overline{m}\rangle$. Since every element of $\mathbb{Z}/n\mathbb{Z}$ is an integer multiple of $\overline{1}$, it follows that $\langle \overline{m}\rangle = \mathbb{Z}/n\mathbb{Z}$.

(3) An isometry $f : D \to D$, where $D \subseteq \mathbb{R}$, preserves distances: $|f(x) - f(y)| = |x - y|$ for all $x, y \in D$. Hence, for $x < y$, we must have $f(y) - f(x) = y - x$ for all pairs, or $f(y) - f(x) = -(y - x)$ for all pairs, so f is either increasing or decreasing. Therefore, $f(x) - x$ is constant in the first case, giving $f(x) = x + c$, while $f(x) + x$ is constant in the second, giving $f(x) = -x + c$. Thus every isometry of a suitable subset of $\mathbb{R}$ has one of these two forms.

(4) Stirling's approximation gives the asymptotic behaviour of factorials:

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \qquad n \to \infty. \tag{94}$$

Equivalently

$$\log(n!) = n \log n - n + \frac{1}{2}\log(2\pi n) + o(1). \tag{95}$$

(5) Thomae's function $T : [0, 1] \to \mathbb{R}$ is defined by

$$T(x) = \begin{cases} 1/q, & x = p/q \in \mathbb{Q} \text{ in lowest terms}, \\ 0, & x \notin \mathbb{Q}. \end{cases} \tag{96}$$

The value at a rational depends on the denominator in its reduced form, while T vanishes at every irrational. It is continuous at every irrational point and discontinuous at every rational point. The key idea is that, in any bounded interval, only finitely many rationals have denominator below a fixed bound.

(6) A subset $A \subseteq \mathbb{R}$ is *dense in* $\mathbb{R}$ if $\overline{A} = \mathbb{R}$.
Equivalently, for every $x \in \mathbb{R}$ and every $\varepsilon > 0$, $(x - \varepsilon, x + \varepsilon) \cap A \neq \varnothing$.
Equivalently, every nonempty open interval in $\mathbb{R}$ contains an element of A.

(7) The Baire Category Theorem states that if X is a complete metric space, then X cannot be expressed as a countable union of nowhere dense subsets.
Equivalently, if $X = \bigcup_{n=1}^{\infty} F_n$ with each F_n closed, then at least one F_n has nonempty interior.
Equivalently, if $U_n \subseteq X$ are dense and open for every n, then $\bigcap_{n=1}^{\infty} U_n$ is dense in X.

16th Aug, 2026. Some mathematics which I encountered on thsi day has been listed below.

(1) For a vector space V and a subspace U, the quotient V/U consists of cosets $v + U$. To show that operations on V/U are well-defined, one checks that changing representatives of the cosets does not change the resulting coset. This follows from U being a subspace.

(2) The span of $S = \{v_1, \ldots, v_n\}$ is the set of all linear combinations $\sum_{i=1}^{n} \lambda_i v_i$. The set S spans V if every vector in V can be expressed as such a linear combination.

(3) Vectors $v_1, \ldots, v_n$ are linearly independent if $\sum_{i=1}^{n} \lambda_i v_i = 0$ implies $\lambda_1 = \cdots = \lambda_n = 0$. If there is a non-trivial linear relation, the vectors are linearly dependent.

(4) A basis of a vector space is a set that is both linearly independent and spanning. Thus, both properties must be checked for a proposed basis.
(5) The standard basis of F^n is $\{e_1, \ldots, e_n\}$, where e_i has 1 in the ith coordinate and 0 elsewhere. Every $(a_1, \ldots, a_n) \in F^n$ can be written as $\sum_{i=1}^{n} a_i e_i$.
(6) The basis of a vector space depends on the field of scalars. For example, $\mathbb{C}$ has basis $\{1\}$ over $\mathbb{C}$ but basis $\{1, i\}$ over $\mathbb{R}$.
(7) The polynomial space $\mathbb{R}[x]$ has basis $\{1, x, x^2, \ldots\}$. Every polynomial is a finite linear combination of these monomials, and uniqueness of polynomial coefficients gives linear independence.
(8) A function $f : X \to F$ has finite support if $\text{supp}(f) = \{x \in X : f(x) \neq 0\}$ is finite.
(9) For each $x \in X$, the function δ_x is 1 at x and 0 elsewhere. The set $\{\delta_x : x \in X\}$ is linearly independent since evaluating a finite linear combination at x_j isolates the coefficient of δ_{x_j}.
(10) If a finite spanning set is linearly dependent, one of its vectors can be expressed as a linear combination of the others and removed without changing the span. Repeating this process gives a linearly independent spanning set, hence a basis.
(11) For a group G, right multiplication $R_g(x) = xg$ gives an anti-homomorphism under the usual left-action convention, while $R_g(x) = xg^{-1}$ gives a genuine left group action.

15th Aug, 2026. This day was mostly spent in trying to understand and revist some topics in mathematics (from varied fields). I am summarising some of them below.

(1) For a polynomial p, the equation $p(x) = a$ has only finitely many real solutions unless p is identically a. These finitely many roots divide $\mathbb{R}$ into finitely many intervals on which $p(x) - a$ has a fixed sign.
(2) A space X is connected if it cannot be written as the union of two disjoint nonempty open subsets. An important result is that the continuous image of a connected space is connected.
(3) If a set can be written as $S = \{\gamma(t) : t \in \mathbb{R}\}$ for a continuous map $\gamma : \mathbb{R} \to \mathbb{R}^n$, then S is connected, since $\mathbb{R}$ itself is connected.
(4) When $a^m = b^n,\ \gcd(m, n) = 1$, the coprimality allows a and b to be expressed as powers of a common quantity: $a = t^n,\ b = t^m$.
This is useful for turning an algebraic relation into a one-parameter parametrization.
(5) If X is a finite set with n elements, then every function $X \to \mathbb{R}$ is continuous when X has the discrete topology. Hence $C(X, \mathbb{R}) \cong \mathbb{R}^n$, so its dimension is n.
(6) The divisors of an integer m usually occur in pairs $d, m/d$. The only time a divisor is paired with itself is when $d = m/d$, which gives $m = d^2$. Thus an integer has an odd number of divisors exactly when it is a perfect square.
(7) Double counting means constructing a set of objects and counting that same set in two different ways. Since both counts give the cardinality of the same set, they must be equal.

(8) For a matrix A, if $Av = \lambda v$, then $A^k v = \lambda^k v$. Consequently, for a convergent power series f, $f(A)v = f(\lambda)v$, so the eigenvalues of $f(A)$ are obtained by applying f to the eigenvalues of A.
(9) For a sequence of functions $f_n : X \to \mathbb{R}$, we say that f_n converges pointwise to $f : X \to \mathbb{R}$ if $\forall x \in X,\ \lim_{n\to\infty} f_n(x) = f(x)$.
Equivalently,$\forall x \in X,\ \forall \varepsilon > 0,\ \exists N = N(x, \varepsilon)$ such that for all $n \geq N$, $|f_n(x) - f(x)| < \varepsilon$.
(10) Uniform convergence $f_n \to f$ means $\sup_x |f_n(x) - f(x)| \to 0$. The important distinction from pointwise convergence is that the same bound works for every x. Therefore, uniform convergence can still be used when the point $x = x_n$ depends on n.
(11) For an operator such as $L(f) = 3f + 2f'$, differentiation lowers the degree of a polynomial by one. This allows the coefficients of a desired preimage to be determined recursively from the highest-degree coefficient downwards, which can establish surjectivity.
(12) Two permutations in S_n are conjugate exactly when they have the same cycle type. If $\sigma^3 = e$, every cycle of σ has length 1 or 3, so only finitely many cycle types are possible.
(13) For a matrix A, the infinity norm is $\|A\|_\infty = \max_i \sum_j |a_{ij}|$. Every eigenvalue satisfies $|\lambda| \leq \|A\|_\infty$. Hence if A has nonnegative entries and every row sums to 1, then every eigenvalue has absolute value at most 1.

14th Aug, 2026. Well, I have not been able to find any naturally arising and interesting example of a presheaf which satisfies the uniqueness axiom for sheaves but satisfies the gluability axiom. There is one example where you take the cover of an open set in a topological space X and define the functions to be, say 0 for all $i \in I$, then this would certainly obey the gluability axiom but separability would cause an issue[2].

Also, for a bipartite graph $K_{m,n}$, the automorphism group is isomorphic to the direct product of the symmetric group, $Aut(K_{m,n}) \equiv S_m \times S_n$. But the questions arises: why the direct product?

Moreover, I had a discussion introducing me to the world to Toposes.

13th Aug, 2026. In algebaric geometry, given an exact sequence of presheaves

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \cdots \longrightarrow \mathcal{F}_n \longrightarrow 0,$$

we show that the following sequence

$$0 \longrightarrow \mathcal{F}_1 U \longrightarrow \mathcal{F}_2 U \longrightarrow \cdots \longrightarrow \mathcal{F}_n U \longrightarrow 0,$$

[2]Please refer to this link for a detailed discussion of this example.

is also exact. Well, this is an iff result and given the definition of a presheaf, which is defined locally, the first exact sequence gives us the second one. We work the other way around by, again, using the definition of presheaves.

12th Aug, 2026. Mathematical ventures on this day included some more reading into scheme-theoretic algebraic notions from Eisenbud, along with some revisitation of linear algebra, particularly diagonal matrices and diagonalizability, and also a result in metric spaces.

A diagonal matrix is a matrix whose entries away from the main diagonal are all zero, i.e.

$$D = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}.$$

It is interesting that a matrix A is diagonalizable precisely when there is a basis consisting of eigenvectors of A, equivalently when

$$P^{-1}AP = D$$

for some invertible matrix P and diagonal matrix D.

The columns of P are simply the linearly independent eigenvectors of A. If

$$Av_i = \lambda_i v_i, \qquad i = 1, \ldots, n,$$

then we take

$$P = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}, \qquad D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Since the eigenvectors are linearly independent, P is invertible, and from

$$AP = PD$$

we obtain

$$P^{-1}AP = D.$$

So, essentially, determining P amounts to finding a basis of eigenvectors of A and placing these eigenvectors as the columns of P.

In metric spaces, every finite metric space with n points can be isometrically embedded into $\mathbb{R}^{n-1}$ with the usual Euclidean metric. It is quite interesting to see an abstract finite metric space being realised concretely as a configuration of points in Euclidean space.

Alongside this, $\mathrm{Spec}(R)$ both as the set of prime ideals of R and through the geometric structure given by the Zariski topology is quite interesting.

We can think of

$$\mathrm{Spec}(R) = \{\mathfrak{p} \subseteq R : \mathfrak{p} \text{ is prime}\}$$

as a collection of prime ideals, and also as a geometric object by putting the Zariski topology on it. For an ideal $I \subseteq R$, we consider

$$V(I) = \{\mathfrak{p} \in \mathrm{Spec}(R) : I \subseteq \mathfrak{p}\},$$

and these sets form the closed sets of the Zariski topology. There is also a structure sheaf $\mathcal{O}_{\mathrm{Spec}(R)}$, and together with the underlying topological space, this gives us the affine scheme associated to R.

10th Aug, 2026. Mathematics on this day included some interesting results about groups acting on metric spaces. One such result was that if a group G acts freely by isometries on $\mathbb{R}^n$, equipped with the Euclidean metric, then G must be torsion-free. This is quite interesting since the geometry of the space places an algebraic restriction on the group itself.

In particular, the idea of algebraic sets as zero sets of ideals and the way varieties arise from imposing irreducibility on these sets is interesting.

09th Aug, 2026. Mathematical ventures on this day included some revisits to linear algebra and analysis, for the most part. The theory of matrices is very rich and gives us some very strong results. For instance, the eigenvalues of a matrix whose entries are non-negative and whose row sums are all equal to 1 must all have absolute value at most 1. That is, if $A = a_{ij}$ satisfies

$$a_{ij} \geq 0, \qquad \sum_j a_{ij} = 1,$$

then every eigenvalue λ of A satisfies

$$|\lambda| \leq 1.$$

In analysis, the Bolzano-Weierstrass theorem tells us that every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence, extending that the *Arzelà-Ascoli* theorem tells us that under suitable conditions, a sequence of functions has a *uniformly* convergent subsequence. In particular, *a uniformly bounded and equicontinuous family of continuous functions on a compact space is relatively compact in the uniform topology.*

Also the idea that a finite metric space can be isometrically embedded into some Euclidean space $\mathbb{R}^n$ is particularly interesting. Thus, given a finite metric space X, there exists some n and a distance-preserving map

$$f : X \longrightarrow \mathbb{R}^n.$$

08th Aug, 2026. Mathematics on this day included some readings on the realisation of groups as metric spaces. We start by taking a group G and its generating set S. Now, for any two elements $g_1, g_2 \in G$ we define the *distance* between them to be the word length $g_1^{-1}g_2$ with respect to S. This induces a metric on G, refered to as the *word metric*. Well, to be precise, the word metric is defined as

$$d_S(g_1, g_2) = |g_1^{-1}g_2|_S,$$

where $|g_1^{-1}g_2|_S$ denotes the minimum number of generators from $S \cup S^{-1}$ needed to express $g_1^{-1}g_2$. Moreover, when we think of groups in terms of their Cayley graaphs, this word metric is precisely the *path metric* of the graph.

Another interesting thing about viewing groups as metric spaces is the analogue of isomorphisms which we obtain in the latter, called an *isometry*. An isometry is simply a bijective function, say f defined as:

$$f : (X, d_X) \to (Y, d_Y),$$

such that

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2)$$

for all pairs $x_1, x_2 \in X$.

Another interesting notion is that of the group of symmetries of a metric space X, denoted as $\mathrm{Isom}(X)$. And a group action of a group G on a metric space X would then be the following group homomorphism

$$G \to \mathrm{Isom}(X).$$

So essentially, for each element $g \in G$, the map gives us an isometry $X \to X$.

07th Aug, 2026. Some mathematics discussions[3] on this day included an alternate proof of the *Bolzano–Weierstrass theorem*. The idea was to start with a bounded sequence $(a_n) \subseteq [a, b]$, divide the interval into two parts, and keep the part which contains infinitely many terms of the sequence. One can keep doing this, obtaining nested intervals

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

whose lengths tend to zero. One can then choose a subsequence by taking a term from each of these intervals, and this subsequence will converge to the unique point determined by the nested intervals. I particularly liked the rather physical analogy used to explain this.

Also, the *Archimedean property* can be realised through the Tortoise and Hare story, and the *Monotone Convergence Theorem* can be visualised by repeatedly partitioning a bounded interval and narrowing down towards a certain point.

[3]Please refer to this webpage and the notes provided there for these ideas.

In another news, an algebraic set can be defined using an ideal rather than a collection of polynomials. Given an ideal $\mathfrak{I} \subseteq k[x_1, \ldots, x_n]$, we can consider

$$V(\mathfrak{I}) = \{a \in \mathbb{A}_k^n : f(a) = 0 \text{ for every } f \in \mathfrak{I}\}$$

.Moreover, the radical of an ideal has some interesting properties that one can play around with in commutative algebra. In particular,

$$V(\mathfrak{I}) = V(\sqrt{\mathfrak{I}}),$$

which is quite interesting since the algebraic set cannot distinguish between an ideal and its radical.

Modules, on the other hand, generally have two ways of being viewed. While I prefer the ring-action approach,

$$R \times M \longrightarrow M,$$

I hope to learn to appreciate the linear algebra approach as well, where modules over rings are considered analogous to vector spaces over a field.

05th Aug, 2026. One of the more surprising ideas I encountered today was that the common notion of the zero set of a collection of polynomials is elevated to a central geometric object in algebraic geometry. Such a set is called an *algebraic set*, while a *variety* is, roughly speaking, an irreducible algebraic set (depending on the convention being used).

I also found it interesting that these objects are studied in affine space ($\mathbb{A}_k^n$). Although affine space has points with coordinates such as $(0, 0)$ or, more generally, $(0, \ldots, 0)$, it is not regarded as having a distinguished origin in the geometric sense. The point $(0, \ldots, 0)$ is then simply another point of the space, and whether it belongs to the zero set of a polynomial depends entirely on the polynomial itself. For instance, if the constant term is nonzero, then $(0, \ldots, 0)$ need not be a zero of that polynomial.

Another pleasing observation was that the Cayley graph of the free group F_2, with respect to its standard free generators, is the infinite 4-regular tree T_4. Starting from any vertex, the tree branches outward in a remarkably symmetric fashion, and after the initial four edges, there are $4 \cdot 3^{n-1}$ new vertices at distance n from the starting point.

Moreover, algebraic sets are interesting objects. Although they can be defined over any field, many of their fundamental properties are studied over algebraically closed fields. There is also the *Zariski topology*, which arises naturally on affine spaces and whose closed sets are precisely the algebraic sets, at least for $n = 1$ and $n = 2$.

04th Aug, 2026. Well, the Cayley graph of a free group with respect to a free basis is an infinite regular tree. This graph has some interesting valencies at each point and it is unlike the finite groups' graphs which contain cycles. In other news, $SL_2(\mathbb{Z})$ has some interesting elements which are also its generators. It would be interesting to explore the geometry of its Cayley graph.

03rd Aug, 2026. Today I learned that automorphisms of a group induce automorphisms of its Cayley graph (provided the generating set is preserved). This is interesting because every automorphism of a group is a permutation of its underlying set, so there is a natural embedding $\mathrm{Aut}(G) \hookrightarrow \mathrm{Sym}(G)$. Thus, the same automorphism acts both as a permutation of the group and as a symmetry of its Cayley graph.